

Tour de Maths 2024

Leg 3

Worked Memorandum



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$\pi = 3, 14159\ 26535\ 89793\ 23846\ 26433\ 83279\ 50288\ 41971\ 69399\ 37510\ 58209\ 74944\ 59230\ 78164\ 06286\ 20899\ 86280\ 34825\ 34211\ 70679\ 82148\ 08651\ 32823\ 06647\ 09384\ 46095\ 50582\ 23172\ 53594\ 08128\ 48111\ 74502\ 84102\ 70193\ 85211\ 05559\ 64462\ 29489\ 54930\ 38196\ 44288\ 10975\ 66593\ 34461\ 28475\ 64823\ 37867\ 83165\ 27120\ 19091\ 45648\ 56692\ 34603\ 48610\ 45432\ 66482\ 13393\ 60726\ 02491\ 41273\ 72458\ 70066\ 06315\ 58817\ 48815\ 20920\ 96282\ 92540\ 91715\ 36436\ 78925\ 90360\ 01133\ 05305\ 48820\ 46652\ 13841\ 46951\ 94151\ 16094\ 33057\ 27036\ 57595\ 91953\ 09218\ 61173\ \dots$

QUESTION 1

What is the probability that a 3 will occur within the first 50 decimal places of π . This excludes the leading 3.

A: 10% B: 14% C: 16% D: 18%

$$\frac{8}{50} = 16\%$$

QUESTION 2

Some students use $\pi = \frac{22}{7}$ in their calculations.

As a percentage, how accurate is this estimation?

A: 4% B: 0,4% C: 0,04% D: 0,004%

$$\frac{\frac{3,14159}{\frac{22}{7}}}{\frac{22}{7}} \cdot 100 = 99,9597\% \text{ or } 0,04\% \text{ off. (wording should have been changed)}$$

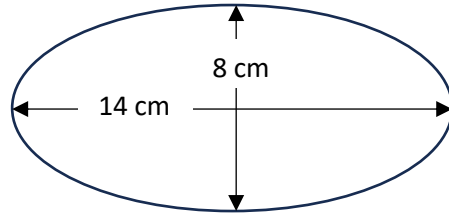
QUESTION 3

The area of an ellipse is given as $A = \pi ab$ where height is $2a$ and length is $2b$.

In this ellipse the height is 8 cm and the length 14 cm.

Calculate the area of the ellipse alongside

In terms of π .



A: 22π

B: 28π

C: 30π

D: 36π

$$A = (4)(7)\pi = 28\pi$$

QUESTION 4

At midnight a frog falls into a hole that is 14,5 meters deep. It can jump 2m upwards but slides back half a meter each time as it rests before the next jump.

How many jumps does it take to get out of the hole?



A: 10 jumps

B: 11 jumps

C: 12 jumps

D: 13 jumps

8 jumps = $8(2 - 0,5) = 12$... still to little for the last jump

9 jumps = $9(2 - 0,5) = 13,5$ and the last jump gets froggie out, therefore 10 jumps

QUESTION 5

An abundant number is defined as a number where the sum of all the positive **proper** factors is greater than the number itself. The proper factors exclude the number and one.

Example:

$F_{22} = 1; 2; 11; 22$ The sum of proper factors: $2 + 11 = 13$. 22 is not abundant.

$F_{12} = 1; 2; 3; 4; 6; 12$ The sum of proper factors: $2 + 3 + 4 + 6 = 15$ 12 is abundant.

What is the smallest abundant number greater than 24?

A: 26

B: 28

C: 30

D: 36

$F_{25} = 1; 5; 25$ X

$F_{28} = 1; 2; 4; 7; 14; 28$ X

$F_{26} = 1; 2; 13; 26$ X

$F_{29} = 1; 29$ X

$F_{27} = 1; 3; 9; 27$ X

$F_{30} = 1; 2; 3; 5; 6; 10; 15; 30$ ✓

QUESTION 6

A six-digit number is written as: 839A6B where A and B are unknown digits.

The number is divisible by 11 and 4. What is the greatest possible value of A+B?

A: 8

B: 9

C: 10

D: 11

$$8 + 3 + 9 + A + 6 + B = 11n$$

$$A + B + 26 = 11n$$

$$\therefore \text{possible pairings } (A; B) = (0; 5)(1; 4)(2; 3)(3; 2)(4; 1)(5; 0)$$

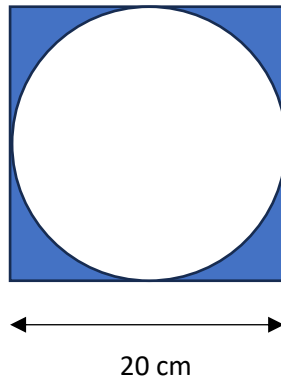
$$\text{And } 6B = 4n$$

$$\therefore B = 0 \text{ or } 4 \text{ or } 8$$

Pairs that are valid are: (5; 0) and (1; 4) therefore greatest $A + B = 9$

QUESTION 7

The figure is a circle inscribed inside a square. Give the area of the shaded region correct to two decimal places.



- A: 40,84 B: 75,00 C: 80,12 D: 85,84

$$20^2 - 10^2\pi = 85,84.$$

QUESTION 8

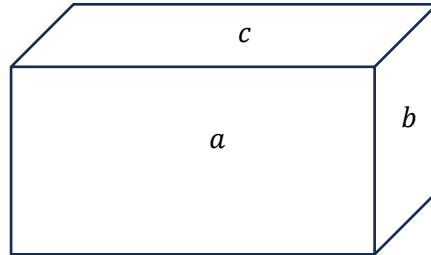
How many non-congruent right angles triangles with integer sides exist, where the perimeter is less than 75.

- A: 7 B: 9 C: 11 D: 14

3, 4, 5 6, 8, 10 9, 12, 15 12, 16, 20 15, 20, 25, 18, 24, 30
5, 12, 13 10, 24, 26 **7, 24, 25** **8, 15, 17,** **20, 21, 29**

QUESTION 9

Three faces of a prism have areas a , b , and c as shown. What is the volume of the box?



- A: $2(a + b + c)$ B: $\sqrt{abc}$ C: $\sqrt{a^2 + b^2 + c^2}$ D: $(a + b + c)$

Label side lengths x, y and z . Areas will be: $a = xz, b = yz$ and $c = xy$.

$$abc = x^2 y^2 z^2 \text{ so } xyz = \sqrt{abc}$$

QUESTION 10

What is the last digit (the units digit) of: $2^{2024} \cdot 3^{2024} \cdot 5^{2024}$?

- A: 0 B: 2 C: 5 D: 7

$2^n \cdot 5^n$ will be a multiple of 10 therefore ends in a zero.

QUESTION 11

Decimal Time: Decimal time is convenient for mathematical calculations, but often needs to be converted back to hours and minutes for everyday usage. Decimal time format is also used to compare time durations in a decimal number.

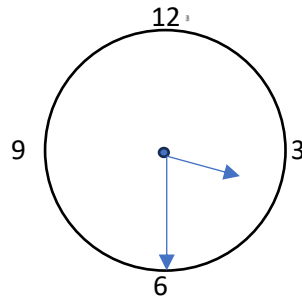
For example, 12:30:15 is represented as 12,5025. One hour is equal to 1,0 and one decimal minute is equal to $1/60$ and one second is equal to $1/3600$.

An event started at 10 minutes past 5 in the evening. What is this in decimal time?

$$17 \text{ hours plus } 10/60 \text{ minutes} = 17,17$$

QUESTION 12

At half-past three, what is the angles formed between the minute hand and the hour hand of a clock?



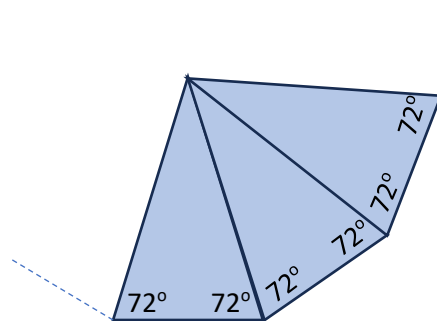
Each hour is $\frac{360^\circ}{12} = 30^\circ$.

Hour hand is halfway between the 3 and the 4, $2 \cdot 30^\circ + 15^\circ = 75^\circ$

QUESTION 13

A series of isosceles triangles all with the equal angles of 72° (as shown) are placed side by side. More triangles are added to make a complete polygon with no overlaps.

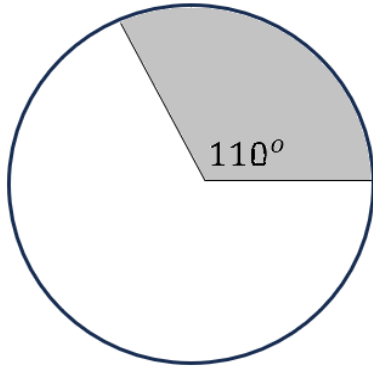
What is the number of sides of the polygon?



Interior angles will each be 36° . 10 triangles will make 360° ... 10 sided polygon.

QUESTION 14

Find the radius of the circle if the area of the shaded region is 50π units².



$$\frac{110^\circ}{360^\circ} \cdot \pi r^2 = 50\pi$$

$$r = \frac{30\sqrt{22}}{11} = 12,79$$

QUESTION 15

Tetration is a cool way of expression numbers in a 'power tower'. ${}^35 = 5^{5^5}$.

Meaning 5 to the power 5 to the power 5.

What is the value of ${}^33 \times {}^42$? Leave your answer in exponential form.

$$3^{27} \cdot 2^{16}$$

QUESTION 16

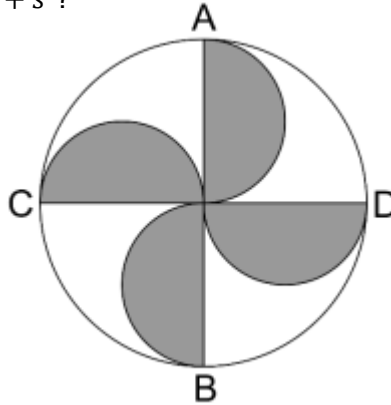
There is a total of 45 boys and girls in a choir. The mean age of the 18 boys is 16,2 years. The mean age of the 27 girls is 16,7 years.

Calculate the mean age of all 45 boys and girls.

$$\frac{18 \cdot 16,2 + 27 \cdot 16,7}{45} = 16,5$$

QUESTION 17

In the circle shown, diameters AB and CD are perpendicular, and each of the four shaded regions is a semicircle. If the ratio of the unshaded area to the shaded area is $\frac{u}{s}$ in simplest form, then what is the value of $u + s$?



$$A_s = 2 \cdot \pi r^2 \quad A_u = A_{big} - A_{small} = \pi(2r)^2 - 2 \cdot \pi r^2 = 4 \cdot \pi r^2 - 2 \cdot \pi r^2 = 2 \cdot \pi r^2$$

$$\frac{u}{s} = \frac{2}{2} = \frac{1}{1} \quad \text{Therefore} \quad u + s = 2$$

QUESTION 18

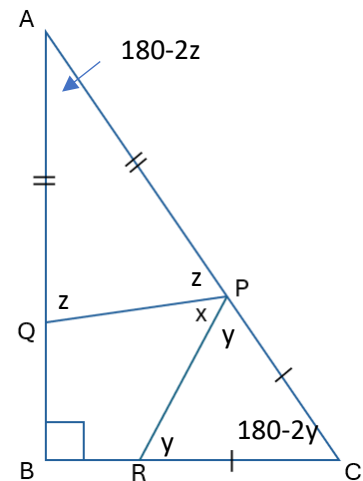
Label angles as shown:

$$180^\circ - 2z + 180^\circ - 2y + 90^\circ = 180^\circ \quad (\text{L's in triangle})$$

$$2z + 2y = 270^\circ$$

$$\therefore z + y = 135^\circ$$

$$x = 45^\circ \quad (\text{L's str line})$$



ABC is a right-angled triangle, and points P, Q, and R are chosen on the sides so that

$AP = AQ$ and $CP = CR$. The diagram is not drawn to scale.

What is the value of x in degrees?

QUESTION 19

A satellite orbits the earth 400km above the earth at a speed of 28 000 km/h. The earth is assumed to be spherical with a circumference of 40 075 km.

How long is one orbit in minutes?



$$r_e = 6378,13$$

$$r_{orbit} = r_e + 400 = 6778,13$$

$$c_{orbit} = 2\pi(6778,13) = 42\,588,27$$

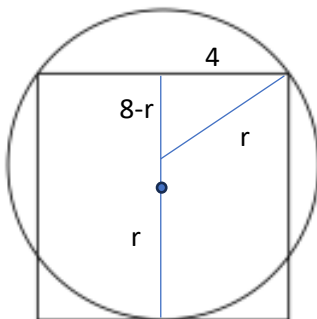
$$t = \frac{42\,588,27}{28\,000} = 1,52hrs = 91\,mins$$

QUESTION 20

Insert constructions and measurements as shown below.

$$4^2 + (8 - r)^2 = r^2 \quad \text{Pythag}$$

$$r = 5$$



In the figure the square with side length 8 has two vertices on the circle, and one side touching the circle.

What is the length of the radius of the circle?

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